Properties of Fixed Effects Estimator in Higher Dimensional Panel

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Abstract:

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1 Introduction

In terms of notation, let x_{ijt} denotes variable over indexes i, j and t then $\bar{x}_{\circ jt} = N_1^{-1}\sum_i x_{ijt}$ and $\bar{x}_{\circ j\circ} = (N_1 T)^{-1}\sum_{i,t} x_{ijt}$. This notation extends naturally to other (combinations) of indexes.

2 Model Setup

This paper concerns with the following model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \mathbf{W}\boldsymbol{\delta} + \boldsymbol{\varepsilon} \tag{1}$$

where $\mathbf{y}$, $\mathbf{X}$ and $\boldsymbol{\beta}_0$ are $N \times 1$ vector, $N \times K$ matrix and $K \times 1$ vector, respectively while $\boldsymbol{\varepsilon}$ is a $N \times 1$ vector of random variables such that $\boldsymbol{\varepsilon} \sim D(\mathbf{0}, \sigma_\epsilon^2\mathbf{I})$ and $\boldsymbol{\varepsilon} \perp \mathbf{X}$. $\mathbf{W}$ and $\boldsymbol{\delta}$ are $N \times K_1$ matrix and $K_1 \times 1$ vector, respectively.

In practice, only $\mathbf{y}$ and $\mathbf{X}$ are observed and the main objective is to obtain consistent estimate of $\boldsymbol{\beta}_0$ under different assumptions of $\mathbf{W}$ and $\boldsymbol{\delta}$. Specifically, this paper focus on $\mathbf{W}\boldsymbol{\delta} = \mathbf{S}_\alpha\boldsymbol{\alpha} + \mathbf{S}_\gamma\boldsymbol{\gamma} + \mathbf{S}_\lambda\boldsymbol{\lambda}$ and consider different assumptions on each of the three components namely, $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$. This generalises the model considered in Balázsi et al. (2024) and Balázsi et al. (2024).

This paper is particularly interested in the estimator of the following form

$$\hat{\boldsymbol{\beta}}(\mathbf{Q}) = \left(\mathbf{X}'\mathbf{Q}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Q}\mathbf{y} \tag{2}$$

where $\mathbf{Q}$ is a $K \times K$ matrix such that it can be written as $\mathbf{Q} = \mathbf{R}\mathbf{R}'$ for some matrix $\mathbf{R}$. Note that Equation 2 captures several well known estimators e.g., when $\mathbf{Q} = \mathbf{I}$ then it reduces into the familiar Ordinary Least Squares Estimator. When $\mathbf{Q} = \boldsymbol{\Omega}^{-1}$ for some appropriate variance-covariance matrix $\boldsymbol{\Omega}$, then it reduces to the well known Generalised Least Squares Estimator. The well known Fixed Effects Estimator can also be written in this form i.e., $\mathbf{Q} = \mathbf{M}$ with $\mathbf{M}$ being an appropriate residual maker such that $\mathbf{M}\mathbf{W} = \mathbf{0}$.

The variance-covariance matrix of $\hat{\beta}(\mathbf{Q})$ is

$$\boldsymbol{\Sigma}(\mathbf{Q}) = \left(\mathbf{X}'\mathbf{Q}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{R}\boldsymbol{\Omega}(\mathbf{Q})\mathbf{R}'\mathbf{X}\left(\mathbf{X}'\mathbf{Q}\mathbf{X}\right)^{-1} \tag{3}$$

where $\boldsymbol{\Omega}(\mathbf{Q}) = \mathbb{E}[\mathbf{e}(\mathbf{Q})\mathbf{e}(\mathbf{Q})']$ with $\mathbf{e}(\mathbf{Q}) = \mathbf{R}'\left[\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\mathbf{Q})\right]$.

2.1 Possible Specifications of $\mathbf{W}$

An implicit assumption in the current model specification is that the data has three dimensions. In the most basic case, each of the error components capture the unique characteristics of each indexes i.e., $\boldsymbol{\alpha} = (\alpha_1 \ldots, \alpha_{N_1})'$, $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_{N_2})'$ and $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_T)'$. Conventionally, this is often expressed as $\alpha_i + \gamma_j + \lambda_t$ for $i = 1, \ldots, N_1$,

	Specification
Single index	$\alpha_i + \gamma_j + \lambda_t$
Double index 1	$\alpha_{ij} + \gamma_j + \lambda_t$
Double index 2	$\alpha_{ij} + \gamma_{jt} + \lambda_t$

Table 1: Possible Specifications

$j = 1, \ldots, N_2$ and $t = 1, \ldots, T$. Obviously, this is not the only specification, other possibilities include two indexes in some of the error components, e.g., $\alpha_{ij} + \gamma_j + \lambda_t$ or $\alpha_{ij} + \gamma_{jt} + \lambda_t$. In fact, there are a total of 336 possible specifications. While the number of possibilities seem large, some of them are trivial while the others are not identifiable or can be reduced into simpler case, at least for purposes of estimation and statistical inference. For this reason, the remaining of the paper will focus on the following cases, Will provide further rationale on why these cases and why are other cases not possible/interesting.

3 $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$ have unique index

3.1 Case 1: Conventional Fixed Effects

This is the natural extension of the conventional two-way fixed effects specification and can be presented as

$$y_{ijt} = \mathbf{x}_{ijt}\boldsymbol{\beta} + \alpha_i + \gamma_j + \lambda_t + \varepsilon_t, \qquad i = 1, \ldots, N_1, j = 1, \ldots, N_2, t = 1, \ldots, T. \tag{4}$$

For purpose of analysis, it is more convenience to analyse the model through its matrix representation. Define $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{N_1})'$, $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_{N_2})'$ and $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_T)'$ with $N = N_1 N_2 T$. The vectors $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$ are constant vector with $\mathbf{S}_\alpha = \mathbf{I}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{i}_T$, $\mathbf{S}_\gamma = \mathbf{i}_{N_1} \otimes \mathbf{I}_{N_2} \otimes \mathbf{i}_T$ and $\mathbf{S}_{\boldsymbol{\lambda}} = \mathbf{i}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{I}_T$.

Consider the case where $\mathbf{Q} = \mathbf{I}$ then the estimator as defined in Equation (2) gives the classical OLS estimator.

Proposition 1. *Under the assumptions that each row of* $\mathbf{X}$ *is a realisation of the* $K \times 1$ *random vector* $\mathbf{x} \sim IID(\boldsymbol{\mu}_x, \boldsymbol{\Omega}_x)$, $N_1^{-1}\mathbf{i}'\boldsymbol{\alpha} - \bar{\alpha} = o(1)$, $N_2^{-1}\mathbf{i}'\boldsymbol{\gamma} - \bar{\gamma} = o(1)$ *and* $T^{-1}\mathbf{i}'\boldsymbol{\lambda} - \bar{\lambda} = o(1)$ *where* $\bar{\alpha}$, $\bar{\gamma}$ *and* $\bar{\lambda}$ *are finite constants. The estimator*

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \boldsymbol{\beta}_0 + \left(\bar{\alpha} + \bar{\gamma} + \bar{\lambda}\right) \mathbf{Q}_X^{-1}\boldsymbol{\mu}_x + o_p(1) \tag{5}$$

as $N_1 \to \infty$, $N_2 \to \infty$ *and* $T \to \infty$ *where* $\boldsymbol{\mu}_X = \mathbb{E}(\mathbf{x})$ *and* $\mathbf{Q}_X = \mathbb{E}(\mathbf{X}'\mathbf{X})$.

Proof. See Appendix. □

Without further assumption on $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$, it should be clear that the middle term on the right hand side of Equation (5) above would not vanish asymptotically. Thus $\hat{\boldsymbol{\beta}}(\mathbf{I})$ is not consistent in general. A special case is when $\boldsymbol{\mu}_X = \mathbf{0}$ and this leads to a particularly interesting situation concerning the sub-vector of $\hat{\boldsymbol{\beta}}$ when a column in $\mathbf{X}$ is $\mathbf{i}_N$, i.e., when including an intercept.

Proposition 2. *Let* $\mathbf{X} = (\mathbf{i}, \mathbf{X}_1)$, $\boldsymbol{\beta}_0 = (\beta_0, \boldsymbol{\beta}_1')'$ *with* $\boldsymbol{\beta}_1 = (\beta_1, \ldots, \beta_{K-1})'$, $\hat{\boldsymbol{\beta}} = \left(\hat{\beta}_0, \hat{\boldsymbol{\beta}}_1'\right)'$ *with* $\hat{\boldsymbol{\beta}}_1 = \left(\hat{\beta}_1, \ldots, \hat{\beta}_{K-1}\right)'$. *Under the assumptions of Proposition 1 and define* $\hat{\boldsymbol{\beta}}(\mathbf{I}) = \left(\hat{\beta}_0, \hat{\boldsymbol{\beta}}_1'\right)'$ *then*

$$\begin{aligned}
\hat{\boldsymbol{\beta}}(\mathbf{I}) &= \left(\hat{\beta}_0(\mathbf{M}_{X_1}), \hat{\boldsymbol{\beta}}_1'(\mathbf{M}_1)\right)' \\
\hat{\beta}_0 &= \beta_0 + \left(\bar{\alpha} + \bar{\gamma} + \bar{\lambda}\right)\mathbf{G} + o_p(1) \\
\hat{\boldsymbol{\beta}}_1(\mathbf{M}_1) &= \boldsymbol{\beta}_1 + o_p(1)
\end{aligned}$$

where $\mathbf{G} = 1 - \boldsymbol{\mu}_{X_1}'\mathbf{Q}^{-1}\boldsymbol{\mu}_{X_1}$

Proof. See Appendix. □

In other words, Proposition 2 confirms that the inclusion of an intercept will allow consistent estimation of the slope coefficients, despite the estimate of the intercept is inconsistent, the slope coefficients are often the focus and in such case, OLS is a consistent estimator for the slopes.

Proposition 3. *Define* $\mathbf{M}_k = \mathbf{I} - \mathbf{S}_k\left(\mathbf{S}_k'\mathbf{S}_k\right)^{-1}\mathbf{S}_k$ *for* $k = \alpha, \gamma, \lambda$ *and* $\mathbf{M}_{kl} = \mathbf{M}_k\mathbf{M}_l$ *for* $k, l = \alpha, \gamma, \lambda$, *then under the assumptions of Proposition 1*

$$\begin{aligned}
\hat{\boldsymbol{\beta}}(\mathbf{M}_{\alpha\gamma\lambda}) &= \boldsymbol{\beta}_0 + o_p(1), \quad \boldsymbol{\Sigma}(\mathbf{M}_{\alpha\gamma\lambda}) = \sigma^2\left(\mathbf{X}'\mathbf{M}_{\alpha\gamma\lambda}\mathbf{X}\right)^{-1} \qquad N \to \infty \\
\hat{\boldsymbol{\beta}}(\mathbf{M}_{\gamma\lambda}) &= \boldsymbol{\beta}_0 + o_p(1), \quad \boldsymbol{\Sigma}(\mathbf{M}_{\gamma\lambda}) = \sigma^2\left(\mathbf{X}'\mathbf{M}_{\gamma\lambda}\mathbf{X}\right)^{-1} \qquad N_1 \to \infty \\
\hat{\boldsymbol{\beta}}(\mathbf{M}_{\alpha\lambda}) &= \boldsymbol{\beta}_0 + o_p(1), \quad \boldsymbol{\Sigma}(\mathbf{M}_{\alpha\lambda}) = \sigma^2\left(\mathbf{X}'\mathbf{M}_{\alpha\lambda}\mathbf{X}\right)^{-1} \qquad N_2 \to \infty \\
\hat{\boldsymbol{\beta}}(\mathbf{M}_{\alpha\gamma}) &= \boldsymbol{\beta}_0 + o_p(1), \quad \boldsymbol{\Sigma}(\mathbf{M}_{\alpha\gamma}) = \sigma^2\left(\mathbf{X}'\mathbf{M}_{\alpha\gamma}\mathbf{X}\right)^{-1} \qquad T \to \infty
\end{aligned}$$

Proof. See Appendix. □

Proposition 3 is well known as it covers the consistency property of a standard Fixed Effects estimator but it also covers the case when the fixed effects was undermisspecified. The result is a direct application and extension of Lemma 2.1 in Chan et al. (2024).

In practice, the dimension of each of these residual makers can be large and this may lead to computational difficulties due to limited memory, despite recent advances

Residual Makers	Element-by-element Operations
$\mathbf{M}_\alpha$	$x_{ijt} - \bar{x}_{i\circ\circ}$
$\mathbf{M}_{\alpha\gamma}$	$x_{ijt} - \bar{x}_{i\circ\circ} - \bar{x}_{\circ j\circ} + \bar{x}_{\circ\circ\circ}$
$\mathbf{M}_{\alpha\gamma\lambda}$	$x_{ijt} - \bar{x}_{i\circ\circ} - \bar{x}_{\circ\circ t} - \bar{x}_{\circ j\circ} + 2\bar{x}_{\circ\circ\circ}$

Table 2: Residual Makers for $\alpha_i + \gamma_j + \lambda_t$

in computing technologies. As such, it is often more convenience to transform the data observation-by-observation rather than relying on matrix operation. Table 2 provides the element-by-element transformation equivalent of each of the residual makers. Discuss the estimation of the variance-covariance matrix.

3.2 Case 2: Error Components Model

The second case is also known as *random effects* or *Error Components Model*. In this case, $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$ are realisation of random vectors. The standard assumptions are

Assumption RE 1. $\boldsymbol{\alpha} \sim D(\mathbf{0}, \boldsymbol{\Omega}_\alpha)$, $\boldsymbol{\gamma} \sim D(\mathbf{0}, \boldsymbol{\Omega}_\gamma)$ and $\boldsymbol{\lambda} \sim D(\mathbf{0}, \boldsymbol{\Omega}_\lambda)$.

Assumption RE 2. Each row in $\mathbf{X}$ is a realisation of $\mathbf{x} \sim IID(\boldsymbol{\mu}_x, \boldsymbol{\Omega}_x)$

Assumption RE 3. $\mathbf{x}$, $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$, $\boldsymbol{\lambda}$ and $\boldsymbol{\varepsilon}$ are pairwise independent.

Proposition 4. *Under Assumptions RE 1 - 3, as $N_1 \to \infty$, $N_2 \to \infty$ and $T \to \infty$,*

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \boldsymbol{\beta}_0 + o_p(1).$$

However, if some of the indexes do not go to infinity then

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \boldsymbol{\beta}_0 + \left(\bar{\alpha}_{N_1} + \bar{\gamma}_{N_2} + \bar{\lambda}_T\right) \mathbf{Q}_X^{-1} \boldsymbol{\mu}_X + o_p(1)$$

where $\bar{\alpha}_{N_1} = N_1^{-1}\mathbf{i}'\boldsymbol{\alpha}$, $\bar{\gamma}_{N_2} = N_2^{-1}\mathbf{i}'\boldsymbol{\gamma}$, and $\bar{\lambda}_T = T^{-1}\mathbf{i}'\boldsymbol{\lambda}$. The terms associated with indexes that tend to infinity would vanish asymptotically.

Proof. See Appendix. □

Proposition 5. *Consider Assumptions 1 - 3, with $\boldsymbol{\Omega}_\alpha = \sigma_\alpha^2 \mathbf{I}_{N_1}$, $\boldsymbol{\Omega}_\gamma = \sigma_\gamma^2 \mathbf{I}_{N_2}$ and $\boldsymbol{\Omega}_\lambda = \sigma_\lambda^2 \mathbf{i}_T$. Let $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}(\mathbf{I})$, $\hat{\boldsymbol{\Omega}}_\alpha = \hat{\sigma}_\alpha^2 \mathbf{I}_{N_1}$, $\hat{\boldsymbol{\Omega}}_\gamma = \hat{\sigma}_\gamma^2 \mathbf{I}_{N_2}$ and $\hat{\boldsymbol{\Omega}}_\lambda = \hat{\sigma}_\lambda^2$ such that*

$$\begin{aligned}
\hat{\sigma}_\epsilon^2 =& N^{-1}\boldsymbol{\varepsilon}'\mathbf{M}_{\alpha\gamma\lambda}\boldsymbol{\varepsilon} \\
\hat{\sigma}_\alpha^2 =& N^{-1}\hat{\boldsymbol{\varepsilon}}'\mathbf{M}_{\gamma\lambda}\hat{\boldsymbol{\varepsilon}} - \hat{\sigma}_\epsilon^2 \\
\hat{\sigma}_\gamma^2 =& N^{-1}\hat{\boldsymbol{\varepsilon}}'\mathbf{M}_{\alpha\lambda}\hat{\boldsymbol{\varepsilon}} - \hat{\sigma}_\epsilon^2 \\
\hat{\sigma}_\lambda^2 =& N^{-1}\hat{\boldsymbol{\varepsilon}}'\mathbf{M}_{\alpha\gamma}\hat{\boldsymbol{\varepsilon}} - \hat{\sigma}_\epsilon^2
\end{aligned}$$

then

$$\hat{\mathbf{\Omega}} - \mathbf{\Omega} = o_p(1) \tag{6}$$

where $\mathbf{\Omega}$ *is the variance-covariance matrix of* $\mathbf{u} = \mathbf{S}_\alpha \boldsymbol{\alpha} + \mathbf{S}_\gamma \boldsymbol{\gamma} + \mathbf{S}_\lambda \boldsymbol{\lambda} + \boldsymbol{\varepsilon}$ *and*

$$\hat{\mathbf{\Omega}} = \hat{\sigma}_\alpha^2 \mathbf{S}_\alpha \mathbf{S}_\alpha' + \hat{\sigma}_\gamma^2 \mathbf{S}_\gamma \mathbf{S}_\gamma' + \hat{\sigma}_\lambda^2 \mathbf{S}_\lambda \mathbf{S}_\lambda' + \hat{\sigma}_\epsilon^2 \mathbf{I}_N. \tag{7}$$

Proof. See Appendix. □

Proposition 6. *Under the assumptions of Proposition 5, the Feasible Generalised Least Squares Estimator as defined by* $\hat{\boldsymbol{\beta}}(\hat{\mathbf{\Omega}})$

$$\hat{\boldsymbol{\beta}}(\hat{\mathbf{\Omega}}) - \boldsymbol{\beta}_0 = o_p(1)$$

and

$$\mathbf{\Sigma}(\hat{\mathbf{\Omega}}) = \left(\mathbf{X}'\hat{\mathbf{\Omega}}^{-1}\mathbf{X}\right)^{-1}.$$

3.3 Case 3: Correlated Unobserved Heterogeneity

There are three possible sub-cases here:

Case 1. The regressors are correlated with at least one of the unobserved heterogeneities but the unobserved heterogeneities are pairwise independent.

Case 2. The regressors are uncorrelated with any of the unobserved heterogeneities are correlated among themselves.

Case 3. The regressors are correlated with at least one of the unobserved heterogeneities and they are also correlated among themselves.

Proposition 7. *Under Assumptions RE 1 - 2*

$$\hat{\boldsymbol{\beta}}(\mathbf{M}_{\gamma\lambda}) = \boldsymbol{\beta}_0 + \left(\mathbf{X}'\mathbf{M}_{\gamma\lambda}\mathbf{X}\right)^{-1} \mathbf{X}'\mathbf{S}_\alpha \boldsymbol{\alpha} + o_p(1)$$

An important point to note about Proposition 7 is that $\mathbf{M}_{\gamma\lambda}\mathbf{S}_\alpha\boldsymbol{\alpha} = \mathbf{S}_\alpha\boldsymbol{\alpha}$, under the assumption that $\mathbb{E}(\alpha_i) = 0$ for all $i = 1, \ldots, N_1$. This leads to the follow two corollaries which provide further insight on the consistency of the Fixed Effect Estimator in the presence of under-misspecification.

Corollary 1. *Under the Assumptions of Proposition 7, let* $\boldsymbol{\alpha}$ *be independent to* $\mathbf{x}$ *with* $\alpha_i \sim IID(0, \sigma_\alpha^2)$ $\forall i$, $\mathbb{E}(\boldsymbol{\alpha}\boldsymbol{\gamma}') = \mathbf{\Omega}_{\alpha\gamma}$ *and* $\mathbb{E}(\boldsymbol{\alpha}\boldsymbol{\lambda}') = \mathbf{\Omega}_{\alpha\lambda}$ *then as* $N_1 \to \infty$

$$\hat{\boldsymbol{\beta}}\left(\mathbf{M}_{\gamma\lambda}\right) = \boldsymbol{\beta}_0 + o_p(1).$$

Corollary 2. *Under the Assumptions of Proposition 7, and assume*

$$\mathbb{E}(x_{k,ijt}\alpha_l) = \begin{cases} \rho_{\alpha k} & \text{if } i = l \\ 0 & \text{otherwise} \end{cases}$$

then as $N_1 \to \infty$

$$\hat{\boldsymbol{\beta}}(\mathbf{M}_{\gamma\lambda}) = \boldsymbol{\beta}_0 + \boldsymbol{\Omega}_X^{-1}\mathbf{S}_\alpha\boldsymbol{\rho}_\alpha + o_p(1)$$

where $\boldsymbol{\rho}_\alpha = (\rho_{\alpha 1} \dots \rho_{\alpha K})'$.

4 $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$ have mixed indexes

The specification considered in this section is $\boldsymbol{\alpha} = (\alpha_{11} \dots, \alpha_{1T}, \alpha_{21}, \dots \alpha_{N_1 T})'$ with $\mathbf{S}_\alpha = \mathbf{I}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{I}_T$, while the definition of other terms remained. This corresponds to the following model

$$y_{ijt} = \mathbf{x}_{ijt}\boldsymbol{\beta}_0 + \alpha_{it} + \gamma_j + \lambda_t + \varepsilon_{ijt}, \quad i = 1, \dots, N_1,\ j = 1, \dots, N_2,\ t = 1, \dots, T. \tag{8}$$

4.1 Case 1: Conventional Fixed Effects

Proposition 8. *Under the assumptions that each row of* $\mathbf{X}$ *is a realisation of the* $K \times 1$ *random vector* $\mathbf{x} \sim IID(\boldsymbol{\mu}_x, \boldsymbol{\Omega}_x)$, $(TN_1)^{-1}\mathbf{i}'\boldsymbol{\alpha} - \bar{\alpha} = o(1)$, $N_2^{-1}\mathbf{i}'\boldsymbol{\gamma} - \bar{\gamma} = o(1)$ *and* $T^{-1}\mathbf{i}'\boldsymbol{\lambda} - \bar{\lambda} = o(1)$. *The estimator yields*

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \boldsymbol{\beta}_0 + \left(\bar{\alpha} + \bar{\gamma} + \bar{\lambda}\right)\mathbf{Q}_X^{-1}\boldsymbol{\mu}_x + o_p(1) \tag{9}$$

as $N_1 \to \infty$, $N_2 \to \infty$ *and* $T \to \infty$ *where* $\boldsymbol{\mu}_X = \mathbb{E}(\mathbf{x})$ *and* $\mathbf{Q}_X = \mathbb{E}(\mathbf{X}'\mathbf{X})$.

Proof. See Appendix. □

Perhaps not surprisingly, the proof of Proposition 8 is identical to the proof of Proposition 1. Again, the consistency of OLS in this case depends on $\boldsymbol{\mu}_x$, given no realistic assumption can be made on $\bar{\alpha}$, $\bar{\gamma}$ and $\bar{\lambda}$ other than in each case, the sequence of sum converges to a finite number. Under such assumption, $\boldsymbol{\mu}_x = \mathbf{0}$ can assure the consistency of $\hat{\boldsymbol{\beta}}(\mathbf{I})$ and therefore, Proposition 2 also applies in this case namely, OLS is consistent if the model contains an intercept i.e., a column of $\mathbf{X}$ is a column of 1s.

Proposition 9. *Under the Assumption of Proposition 8, as* $N_1 \to \infty$, $N_2 \to \infty$ *and* $T \to \infty$ *then*

$$\hat{\boldsymbol{\beta}}\left(\mathbf{M}_\alpha\right) = \boldsymbol{\beta}_0 + o_p(1).$$

Proof. See Appendix. □

Proposition 10. *Under the assumptions that each row of* $\mathbf{X}$ *is a realisation of the* $K \times 1$ *random vector* $\mathbf{x} \sim IID(\boldsymbol{\mu}_x, \boldsymbol{\Omega}_x)$*, as* $N \to \infty$ *with fixed* N_2*,*

$$\hat{\boldsymbol{\beta}}(\mathbf{M}_\alpha) = \boldsymbol{\beta}_0 + o_p(1).$$

Proof. See Appendix. □

Proposition 10 provides a result without assuming the convergence of the sequence of sums of any of the fixed effects.

Proposition 11. *Under the assumptions of Proposition 9,*

$$\hat{\boldsymbol{\beta}}\left(\mathbf{M}_{\gamma\lambda}\right) = \boldsymbol{\beta}_0 + \left(\mathbf{X}'\mathbf{M}_{\gamma\lambda}\mathbf{X}\right)^{-1}\mathbf{X}\mathbf{M}_{\gamma\lambda}\mathbf{S}_\alpha\boldsymbol{\alpha} + o_p(1). \tag{10}$$

Proof. See Appendix. □

Note that $\mathbf{M}_{\gamma\lambda}\mathbf{X}$ represents the familiar two ways fixed effects transform i.e., $\tilde{x}_{ijt} = x_{ijt} - \bar{x}_{i\circ t} - \bar{x}_{ij\circ} + \bar{x}$ while $\mathbf{M}_{\gamma\lambda}\mathbf{S}_\alpha\boldsymbol{\alpha}$ gives $\alpha_{it} - \bar{\alpha}_{\circ t}$. Since $\boldsymbol{\alpha}$ is a fixed i.e., constant vector and not random. The middle term is unlikely to vanish in finite sample or asymptotically. Perhaps more importantly, Propositions 9 - 11 collectively imply that, in the case of fixed effects with mixed indexes, if the fixed effect with the mixed indexes is accommodated with the residual maker then the remaining fixed effects with single index are not going to affect the consistency of the estimator.

4.2 Case 2: Error Components Model

This subsection considers the case when the unobserved heterogenieties are random rather than fixed i.e., constant. There are number of possibilities in this case. As a starting point, consider the following assumption:

MRE-1 Assumption 1. $\boldsymbol{\alpha} \sim D(\mathbf{0}, \boldsymbol{\Omega}_\alpha \otimes I_T)$, $\boldsymbol{\gamma} \sim D(\mathbf{0}, \sigma_\gamma^2 I_{N_2})$ and $\boldsymbol{\lambda} = \mathbf{0}$

MRE-1 Assumption 2. $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$ are pairwise independent and are also independent to $\mathbf{X}$ and $\boldsymbol{\varepsilon}$.

Together these assumptions represent one of the simplest case that allows for interdependence across a particular index. Specifically, Assumption 1 allows dependence over index i as described by the variance-covariance matrix $\boldsymbol{\Omega}_\alpha$ which is constant over the t index. The specification of the error components in this case is $\alpha_{it} + \gamma_j + \varepsilon_{ijt}$

Proposition 12. *Under MRE-1 Assumptions 1 - 2, as* $N_2 \to \infty$ *and* $T \to \infty$

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \boldsymbol{\beta}_0 + o_p(1) \tag{11}$$

with

$$\boldsymbol{\Omega}(\mathbf{I}) = \mathbf{S}_\alpha\left(\boldsymbol{\Omega}_\alpha \otimes I_T\right)\mathbf{S}_\alpha' + \sigma_\gamma^2\mathbf{S}_\gamma\mathbf{S}_\gamma' + \sigma_\epsilon^2\mathbf{I}_N. \tag{12}$$

Even though the standard OLS estimator is consistent in this case, the estimation of $\boldsymbol{\Omega}(\mathbf{I})$ may be challenging in practice, given the dependence structure of $\boldsymbol{\alpha}$. In order to estimate $\boldsymbol{\Omega}(\mathbf{I})$ and subsequently $\boldsymbol{\Sigma}(\mathbf{I})$, consider the following procedures:

1. Obtain the estimated residual vector $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\mathbf{I})$.
2. Compute $\mathbf{u} = \mathbf{M}_\alpha \hat{\boldsymbol{\varepsilon}}$ where $\mathbf{M}_\alpha = \mathbf{I} - T^{-1}\left(\mathbf{I}_{N_1} \otimes \mathbf{I}_{N_2} \otimes \mathbf{J}_T'\right)$.
3. Compute $\tilde{\sigma}^2 = N^{-1}\mathbf{u}'\mathbf{u}$.
4. Compute $\mathbf{v} = \mathbf{M}_\gamma \hat{\boldsymbol{\varepsilon}}$.
5. Compute $\hat{\boldsymbol{\Omega}}_\alpha = T^{-1}\sum_{t=1}^{T} \mathbf{v}_{\circ jt}\mathbf{v}_{\circ jt}' - \tilde{\sigma}^2\mathbf{I}$ for any $j = 1, \ldots, N_2$.

MRE-2 Assumption 1. $\boldsymbol{\alpha} \sim D(\mathbf{0}, \boldsymbol{\Omega}_\alpha \otimes I_T)$, $\boldsymbol{\gamma} \sim D(\mathbf{0}, \sigma_\gamma^2 I_{N_2})$ and $\boldsymbol{\lambda} \sim D(\mathbf{0}, \sigma_\lambda^2 I_T)$

MRE-2 Assumption 2. $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$ are pairwise independent and are also independent to $\mathbf{X}$ and $\boldsymbol{\varepsilon}$.

Step 2 can be interpreted as $\tilde{\varepsilon}_t = \varepsilon_{ijt} - \bar{\epsilon}_{ij\circ}$ Assumption 1 consider the case that there are dependence across the i index through the covariance matrix $\boldsymbol{\Omega}_\alpha$ which describe how each observation along the i index are correlated. The correlation, however, is constant over time index but their realizations are fixed at the j index. In other words, the specification of the error components in this case is $\alpha_{it} + \gamma_j + \lambda_t + \varepsilon_{ijt}$.

Proposition 13. *Under MRE-2 Assumptions 1 - 2, as $N_2 \to \infty$ and $T \to \infty$*

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \boldsymbol{\beta}_0 + o_p(1) \tag{13}$$

with

$$\boldsymbol{\Omega}(\mathbf{I}) = \mathbf{S}_\alpha\left(\boldsymbol{\Omega}_\alpha \otimes I_T\right)\mathbf{S}_\alpha' + \sigma_\gamma^2\mathbf{S}_\gamma\mathbf{S}_\gamma' + \sigma_\lambda^2\mathbf{S}_\lambda\mathbf{S}_\lambda' + \sigma_\epsilon^2\mathbf{I}_N. \tag{14}$$

The estimation of $\boldsymbol{\Omega}(\mathbf{I})$ in this case is similar to the case considered in Proposition 12, but individual identifications of $\boldsymbol{\Omega}_\alpha$ and σ_λ^2 may be challenging. However, for purpose of valid statistical inference on $\hat{\beta}$, such identifications are not required, so long as $\boldsymbol{\Omega}(\mathbf{I})$ can be consistently estimated.

To demonstrate, consider

$$\begin{aligned}\mathbf{S}_\alpha\boldsymbol{\alpha} + \mathbf{S}_\lambda\boldsymbol{\lambda} &= \left(\mathbf{I}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{I}_T\right)\boldsymbol{\alpha} + \left(\mathbf{i}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{I}_T\right)\boldsymbol{\lambda} \\ &= \left(\mathbf{I}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{I}_T\right)\left[\boldsymbol{\alpha} + \left(\mathbf{i}_{N_1} \otimes \mathbf{I}_T\right)\boldsymbol{\lambda}\right]\end{aligned}$$

and let $\boldsymbol{\Omega}_{\alpha\lambda}$ denotes the variance-covariance matrix of $\boldsymbol{\alpha} + \left(\mathbf{i}_{N_1} \otimes \mathbf{I}_T\right)\boldsymbol{\lambda}$ then

$$\boldsymbol{\Omega}(\mathbf{I}) = \mathbf{S}_\alpha\boldsymbol{\Omega}_{\alpha\lambda}\mathbf{S}_\alpha' + \sigma_\gamma^2\mathbf{S}_\gamma\mathbf{S}_\gamma'.$$

Since $\boldsymbol{\Omega}_{\alpha\lambda}$ is a constant matrix along the j index, it can be estimated consistently as $N_2 \to \infty$. This implies one can obtain a consistent estimator of $\boldsymbol{\Omega}(\mathbf{I})$ using the steps below:

1. Obtain the estimated residual vector $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\mathbf{I})$.
2. Compute $\mathbf{u} = \mathbf{M}_{\alpha\gamma}\hat{\boldsymbol{\varepsilon}}$ where
$$\begin{aligned}\mathbf{M}_{\alpha\gamma} =& \mathbf{I} - (N_1 T)^{-1} (\mathbf{J}_{N_1} \otimes \mathbf{I}_{N_2} \otimes \mathbf{J}_T) - N_2^{-1} (\mathbf{I}_{N_1} \otimes \mathbf{J}_{N_2} \otimes \mathbf{I}_T) \\ &+ N^{-1} (\mathbf{J}_{N_1} \otimes \mathbf{J}_{N_2} \otimes \mathbf{J}_T)\end{aligned}$$
3. Compute $\tilde{\sigma}^2 = N^{-1}\mathbf{u}'\mathbf{u}$.
4. Compute $\mathbf{w} = \mathbf{M}_{\alpha}\hat{\boldsymbol{\varepsilon}}$.
5. Compute $\hat{\sigma_\gamma} = N^{-1}\mathbf{w}'\mathbf{w} - \tilde{\sigma}^2$.
6. Compute $\mathbf{v} = \mathbf{M}_{\gamma}\hat{\boldsymbol{\varepsilon}}$ where
$$\mathbf{M}_{\gamma} = \mathbf{I} - (N_1 T)^{-1} (\mathbf{J}_{N_1} \otimes \mathbf{I}_{N_2} \otimes \mathbf{J}_T)$$
7. Compute $\hat{\boldsymbol{\Omega}}_{\alpha\lambda} = N_2^{-1} \sum_{j=1}^{N_2} \mathbf{v}_{\circ j \circ}\mathbf{v}'_{\circ j \circ} - \tilde{\sigma}^2\mathbf{I}$.

5 Limitation on Identification

There are cases where identification of $\boldsymbol{\Omega}$ is only possible when all the error components are homoskedastic with no dependencies. One example is

$$y_{ijt} = \mathbf{x}'_{ijt}\boldsymbol{\beta} + \alpha_{it} + \gamma_{jt} + \lambda_t + \varepsilon_{ijt} \quad i = 1, \ldots, N_1, j = 1, \ldots, N_2, t = 1, \ldots, T. \tag{15}$$

Set $\boldsymbol{\alpha} = (\alpha_{11}, \ldots, \alpha_{1T}, \alpha_{21}, \ldots, \alpha_{N_1 T})'$, $\boldsymbol{\gamma} = (\gamma_{11}, \ldots, \gamma_{1T}, \gamma_{21}, \ldots, \gamma_{N_2 T})'$, $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_T)'$ and $\boldsymbol{\varepsilon} = (\varepsilon_{111}, \ldots, \varepsilon_{N_1 N_2, T})'$ then the model has the following representation in the matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{S}_\alpha\boldsymbol{\alpha} + \mathbf{S}_\gamma\boldsymbol{\gamma} + \mathbf{S}_\lambda\boldsymbol{\lambda} + \boldsymbol{\varepsilon} \tag{16}$$

where

$$\begin{aligned}\mathbf{S}_\alpha =& \mathbf{I}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{I}_T \\ \mathbf{S}_\gamma =& \mathbf{i}_{N_1} \otimes \mathbf{I}_{N_2} \otimes \mathbf{I}_T \\ \mathbf{S}_\lambda =& \mathbf{i}_{N_1} \otimes \mathbf{i}_{N_1} \otimes \mathbf{I}_T.\end{aligned}$$

Moreover,

$$\begin{aligned}\boldsymbol{\alpha} \sim& D\left(0, \sigma_\alpha^2 \mathbf{I}_{N_1 T}\right) \\ \boldsymbol{\gamma} \sim& D\left(0, \sigma_\gamma^2 \mathbf{I}_{N_2 T}\right) \\ \boldsymbol{\lambda} \sim& D\left(0, \sigma_\lambda^2 \mathbf{I}_{N_T}\right) \\ \boldsymbol{\varepsilon} \sim& D\left(0, \sigma_\varepsilon^2 \mathbf{I}_N\right)\end{aligned}$$

Moreover, it is also assumed that each element in $\mathbf{x}_{ijt}$ are pairwise independent to elements in $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$, $\boldsymbol{\lambda}$ and $\boldsymbol{\varepsilon}$. The relevant residual makers are

$$\begin{aligned}
\mathbf{M}_\alpha =& \mathbf{I} - N_2^{-1}\left(\mathbf{I}_{N_1} \otimes \mathbf{J}_{N_2} \otimes \mathbf{I}_T\right) \\
\mathbf{M}_\gamma =& \mathbf{I} - N_1^{-1}\left(\mathbf{J}_{N_1} \otimes \mathbf{I}_{N_2} \otimes \mathbf{I}_T\right) \\
\mathbf{M}_{\alpha\gamma} \equiv& \mathbf{M}_\alpha \mathbf{M}_\gamma \\
=& \mathbf{I} - N_2^{-1}\left(\mathbf{I}_{N_1} \otimes \mathbf{J}_{N_2} \otimes \mathbf{I}_T\right) \\
& - N_1^{-1}\left(\mathbf{J}_{N_1} \otimes \mathbf{I}_{N_2} \otimes \mathbf{I}_T\right) \\
& + \left(N_1 N_2\right)^{-1}\left(\mathbf{J}_{N_1} \otimes \mathbf{J}_{N_2} \otimes \mathbf{I}_T\right)
\end{aligned}$$

Note that $\mathbf{M}_\alpha \mathbf{S}_\lambda = \mathbf{M}_\gamma \mathbf{S}_\lambda = \mathbf{0}$. Therefore, it is not possible to eliminate $\boldsymbol{\alpha}$ or $\boldsymbol{\gamma}$ without removing $\boldsymbol{\lambda}$ using the residual makers.

The variance-covariance matrix, $\boldsymbol{\Omega}$ takes the form

$$\sigma_\alpha^2 \mathbf{S}_\alpha \mathbf{S}_\alpha' + \sigma_\gamma^2 \mathbf{S}_\gamma \mathbf{S}_\gamma' + \sigma_\lambda^2 \mathbf{S}_\lambda \mathbf{S}_\lambda' + \sigma_\varepsilon^2 \mathbf{I}. \tag{17}$$

Thus, one approach to obtain a consistent estimate of $\boldsymbol{\Omega}$ is to obtain consistent estimates of σ_α^2, σ_γ^2, σ_λ^2 and σ_ε^2. The process of estimating each of these quantities can be found below

Step 1. Obtain a consistent estimate of $\boldsymbol{\beta}$. Given the independencies between the error components and the regressors, OLS is consistent. Therefore $\hat{\boldsymbol{\beta}}(\mathbf{I})$ i.e., OLS, is a consistent estimator of $\boldsymbol{\beta}$ for this specification.

Step 2. Obtain the estimated residual $\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\mathbf{I})$.

Step 3. Estimate σ_ε^2.

To do this, first obtain $\mathbf{M}_{\alpha\gamma}\hat{\mathbf{u}}$. In the event matrix operation is not feasible, the same can be achieved via the following transformation

$$\hat{u}_{ijt} - \bar{\hat{u}}_{i\circ t} - \bar{\hat{u}}_{\circ jt} + \bar{\hat{u}}_{\circ\circ t}$$

where $\bar{\hat{u}}_{\circ jt}$ denotes the sample mean of $\hat{u}_{ijt}$ over the i index. The definition extends naturally to other (combinations) of indexes. For example, $\bar{\hat{u}}_{\circ\circ t}$ denotes the sample mean of $\hat{u}_{ijt}$ over the i and j indexes. This gives $\mathbf{M}_{\alpha\gamma}\hat{\boldsymbol{\varepsilon}}$. Note that

$$\begin{aligned}
N^{-1}\mathbb{E}\left(\boldsymbol{\varepsilon}'\mathbf{M}_{\alpha\gamma}\boldsymbol{\varepsilon}\right) =& N^{-1}\mathbb{E}\left[\operatorname{tr}\left(\mathbf{M}_{\alpha\gamma}\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\right)\right] \\
=& N^{-1}\sigma_\varepsilon^2 \operatorname{tr}\left(\mathbf{M}_{\alpha\gamma}\right). \\
=& \sigma_\varepsilon^2\left(1 - \frac{1}{N_2} - \frac{1}{N_1} + \frac{1}{N_1 N_2}\right).
\end{aligned}$$

This implies

$$\hat{\sigma}_\varepsilon^2 = \frac{N_1 N_2}{N_1 N_2 - N_1 - N_2 + 1} \boldsymbol{\varepsilon}' \mathbf{M}_{\alpha\gamma} \boldsymbol{\varepsilon} \tag{18}$$

is a consistent estimator for σ_ε^2.

Step 4. Estimate σ_α^2.

To do this, consider $\mathbf{M}_\gamma \hat{\mathbf{u}}$, which is equivalent to the following transformation on $\hat{u}_{ijt}$

$$\hat{u}_{ijt} - \bar{\hat{u}}_{\circ jt}.$$

This gives $\mathbf{M}_\gamma \mathbf{S}_\alpha \hat{\boldsymbol{\alpha}} + \mathbf{M}_\gamma \hat{\boldsymbol{\varepsilon}}$. Thus

$$\begin{aligned} N^{-1}\mathbb{E}\left(\hat{\mathbf{u}}'\mathbf{M}_\gamma \hat{\mathbf{u}}\right) =& N^{-1}\mathbb{E}\left[\hat{\boldsymbol{\alpha}}'\mathbf{S}_\alpha'\mathbf{M}_\gamma \mathbf{S}_\alpha \hat{\boldsymbol{\alpha}} + \hat{\boldsymbol{\varepsilon}}'\mathbf{M}_\gamma \hat{\boldsymbol{\varepsilon}}\right] \\ =& N^{-1}\left(\hat{\sigma}_\alpha^2 \mathrm{tr}\left(\mathbf{S}_\alpha'\mathbf{M}_\gamma \mathbf{S}_\alpha\right) + \hat{\sigma}_\varepsilon^2 \mathrm{tr}\left(\mathbf{M}_\gamma\right)\right) \\ =& \frac{N_1 - 1}{N_1}\hat{\sigma}_\alpha^2 + \frac{N_1 - 1}{N_1}\hat{\sigma}_\varepsilon^2 \end{aligned}$$

Hence

$$\hat{\sigma}_\alpha^2 = \frac{1}{(N_1 - 1)N_2 T}\hat{\mathbf{u}}'\mathbf{M}_\gamma \hat{\mathbf{u}} - \hat{\sigma}_\varepsilon^2. \tag{19}$$

Step 5. Estimate σ_γ^2.

This step follow the exact same argument as previous step by considering $\mathbf{M}_\alpha \hat{\mathbf{u}}$, which is equivalent to

$$\hat{u}_{ijt} - \bar{\hat{u}}_{i\circ t}.$$

This gives

$$\hat{\sigma}_\gamma^2 = \frac{1}{N_1(N_2 - 1)T}\hat{\mathbf{u}}'\mathbf{M}_\alpha \hat{\mathbf{u}} - \hat{\sigma}_\varepsilon^2 \tag{20}$$

Step 6. Estimate σ_λ^2.

Note that

$$\begin{aligned} N^{-1}\mathbb{E}\left(\hat{\mathbf{u}}'\mathbf{u}\right) =& N^{-1}\left(\hat{\sigma}_\alpha^2 \mathrm{tr}\left(\mathbf{S}_\alpha'\mathbf{S}_\alpha\right) + \hat{\sigma}_\gamma^2 \mathrm{tr}\left(\mathbf{S}_\gamma'\mathbf{S}_\gamma\right) + \hat{\sigma}_\lambda^2 \mathrm{tr}\left(\mathbf{S}_\lambda \mathbf{S}_\lambda\right) + \hat{\sigma}_\varepsilon^2\right) \\ =& \hat{\sigma}_\alpha^2 + \hat{\sigma}_\gamma^2 + \hat{\sigma}_\lambda^2 + \hat{\sigma}_\varepsilon^2 \end{aligned}$$

and therefore

$$\hat{\sigma}_\lambda^2 = N^{-1}\hat{\mathbf{u}}'\hat{\mathbf{u}} - \hat{\sigma}_\alpha^2 - \hat{\sigma}_\gamma^2 - \hat{\sigma}_\varepsilon^2 \tag{21}$$

6 $\boldsymbol{\alpha}$ is fixed but $\boldsymbol{\gamma}$ and $\boldsymbol{\lambda}$ are random

7 Monte Carlo Evidence

7.1 Design of Experiments

This section describes the data-generating process (DGP) and the estimators implemented in the Monte Carlo experiments. The objective of the simulations is to evaluate the finite-sample and asymptotic behavior of the estimators $\hat{\boldsymbol{\beta}}(\mathbf{Q})$ under different assumptions on the covariance structure of the composite error term.

7.1.1 Panel Structure and Indexing

We consider a balanced three-dimensional panel indexed by $i = 1, \ldots, N_1$, $j = 1, \ldots, N_2$, and $t = 1, \ldots, T$, with $N = N_1 N_2 T$. Observations are stacked lexicographically into vectors with i varying fastest, followed by j, and then t. This ordering is used consistently when constructing $\mathbf{S}_\alpha$, $\mathbf{S}_\gamma$, and $\mathbf{S}_\lambda$ as well as covariance matrices.

For each configuration (N_1, N_2, T), the covariance matrices of the random components are drawn once and held fixed across Monte Carlo replications. The regressors and error realizations are redrawn in each replication.

7.1.2 Data-Generating Process

The DGP is given by

$$y_{ijt} = c + \beta_0 x_{ijt} + \alpha_i + \gamma_j + \lambda_t + u_{ijt}, \tag{22}$$

or, in stacked matrix notation,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \mathbf{S}_\alpha \boldsymbol{\alpha} + \mathbf{S}_\gamma \boldsymbol{\gamma} + \mathbf{S}_\lambda \boldsymbol{\lambda} + \mathbf{u}, \tag{23}$$

where $\mathbf{X} = (\mathbf{i}_N, \mathbf{x})$ includes an intercept unless stated otherwise.

Regressor and Idiosyncratic Error. The scalar regressor and idiosyncratic error are generated as

$$x_{ijt} \sim \mathcal{N}(\mu_x, \sigma_x^2),$$
$$u_{ijt} \sim \mathcal{N}(0, \sigma_u^2),$$

independently across (i, j, t) and independently of all fixed effects. This ensures $\mathbf{X} \perp (\boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\lambda}, \mathbf{u})$ under the maintained design.

Fixed Effects. The fixed-effect vectors are drawn from multivariate normal distributions,

$$\begin{aligned}\boldsymbol{\alpha} &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}_\alpha),\\ \boldsymbol{\gamma} &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}_\gamma),\\ \boldsymbol{\lambda} &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}_\lambda).\end{aligned}$$

Two covariance specifications are considered:

1. **Diagonal (Homoskedastic) Case:** $\boldsymbol{\Omega}_\alpha = \sigma_\alpha^2 \mathbf{I}_{N_1}$, $\boldsymbol{\Omega}_\gamma = \sigma_\gamma^2 \mathbf{I}_{N_2}$, $\boldsymbol{\Omega}_\lambda = \sigma_\lambda^2 \mathbf{I}_T$.
2. **Full-Block Case:** Each covariance matrix is a symmetric positive definite matrix, allowing arbitrary cross-sectional dependence within the corresponding dimension.

The composite error is therefore

$$\boldsymbol{\varepsilon} = \mathbf{S}_\alpha \boldsymbol{\alpha} + \mathbf{S}_\gamma \boldsymbol{\gamma} + \mathbf{S}_\lambda \boldsymbol{\lambda} + \mathbf{u}, \tag{24}$$

with population covariance matrix

$$\boldsymbol{\Omega} = \mathbf{S}_\alpha \boldsymbol{\Omega}_\alpha \mathbf{S}_\alpha' + \mathbf{S}_\gamma \boldsymbol{\Omega}_\gamma \mathbf{S}_\gamma' + \mathbf{S}_\lambda \boldsymbol{\Omega}_\lambda \mathbf{S}_\lambda' + \sigma_u^2 \mathbf{I}_N. \tag{25}$$

7.1.3 Estimators Implemented

For each Monte Carlo replication, the following estimators are computed on the same simulated dataset.

(i) Pooled OLS. The pooled estimator corresponds to $\mathbf{Q} = \mathbf{I}_N$:

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

This estimator ignores all dependence induced by $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$, and $\boldsymbol{\lambda}$.

(ii) Fixed Effects OLS. Fixed-effects estimators are implemented via within transformation. Element-by-element transformations are used instead of explicitly constructing large residual-maker matrices. For k dimensions removed, the transformation applied to x_{ijt} is

$$\tilde{x}_{ijt} = x_{ijt} + (k-1)\bar{x} - \sum_{d \in D} \bar{x}_d,$$

with the same transformation applied to y_{ijt}. After transformation, $\hat{\boldsymbol{\beta}}$ is obtained by OLS without an intercept.

(iii) FGLS1. FGLS1 is a one-step feasible GLS estimator based on a parametric estimate of the covariance components. The procedure is:

1. Obtain residuals from a fixed-effects OLS regression.
2. Estimate σ_u^2 using a threeway fixed effect regression residuals.
3. Construct estimated covariance matrices $(\hat{\Omega}_\alpha, \hat{\Omega}_\gamma, \hat{\Omega}_\lambda)$ assuming either diagonal or full-block structure, as specified.

   Example: To estimate Ω_α, first run a fixed effect regression with fixed effects for j and t, to get the residuals.

   - **Case 1:** Ω_α is diagonal (homoschedastic case): $\hat{\sigma}_\alpha^2 = \mathrm{Var}[\hat{\varepsilon}_{ijt}] - \hat{\sigma}_u^2$,.
   - **Case 2:** Ω_α is a full block: $\hat{\Omega}_\alpha = \frac{1}{N_2 * T}\, \hat{\varepsilon}_{.jt}\, \hat{\varepsilon}'_{.jt} - I_N * \hat{\sigma}_u^2$.

4. Assemble
$$\hat{\Omega} = S_\alpha\, \hat{\Omega}_\alpha\, S_\alpha + S_\gamma\, \hat{\Omega}_\gamma\, S_\gamma + S_\lambda\, \hat{\Omega}_\lambda\, S_\lambda + \hat{\sigma}_u^2 I_N.$$
5. Compute the GLS estimator
$$\hat{\beta}_{\mathrm{FGLS1}} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}y.$$

(iv) FGLS2. FGLS2 is a one-step feasible GLS estimator that differs from FGLS1 only in the construction of the covariance components. While FGLS1 estimates each variance component from level residuals of auxiliary fixed-effect regressions, FGLS2 relies on *double-difference* transformations that difference out selected fixed effects by construction. This leads to variance estimators that are robust to high-dimensional incidental parameters and avoids the downward bias induced by naive block extraction from global residual outer products.

The procedure is as follows:

1. Obtain residuals $\hat{\varepsilon}_{ijt}$ from a three-way fixed-effects OLS regression (with fixed effects for i, j, and t).
2. Estimate σ_u^2 using double-difference residual contrasts. Specifically, for fixed t consider all 2×2 rectangles in the (i, j) dimension and define
$$d_{ij,i'j',t} = \hat{\varepsilon}_{ijt} - \hat{\varepsilon}_{ij't} - \hat{\varepsilon}_{i'jt} + \hat{\varepsilon}_{i'j't}.$$
   Under the maintained model, $d_{ij,i'j',t}$ differences out α_i, γ_j, and λ_t, leaving
$$\mathrm{Var}(d_{ij,i'j',t}) = 4\sigma_u^2.$$
   The estimator is therefore
$$\hat{\sigma}_u^2 = \frac{1}{4}\,\mathrm{Var}(d),$$
   where the variance is computed across all admissible rectangles and time periods.

3. Construct estimated covariance matrices $(\hat{\Omega}_i, \hat{\Omega}_j, \hat{\Omega}_t)$ using the appropriate residuals, with the specific differencing scheme chosen to eliminate all components except the one of interest.

Example: Estimating Ω_α. To estimate the covariance of α_i, apply a differencing scheme that eliminates γ_j, λ_t, and u_{ijt} in expectation. This is achieved by averaging double differences over (j, t) while keeping i fixed.

- **Case 1: Ω_α diagonal (homoskedastic).**
  Let $\hat{\varepsilon}_{ijt}^{(\gamma,\lambda)}$ denote the residual from the within regression that partials out (γ_j, λ_t). Under the maintained model,
  $$\hat{\varepsilon}_{ijt}^{(\gamma,\lambda)} = \alpha_i + u_{ijt}.$$
  Averaging over (j, t) at fixed i yields
  $$\bar{\varepsilon}_i = \frac{1}{N_2 T} \sum_{j=1}^{N_2} \sum_{t=1}^{T} \hat{\varepsilon}_{ijt}^{(\gamma,\lambda)} = \alpha_i + \bar{u}_i,$$
  where $\mathrm{Var}(\bar{u}_i) = \sigma_u^2/(N_2 T)$.
  A consistent estimator of σ_α^2 is therefore
  $$\hat{\sigma}_\alpha^2 = \mathrm{Var}_i(\bar{\varepsilon}_i) - \frac{\hat{\sigma}_u^2}{N_2 T},$$
  and the corresponding covariance estimator is
  $$\hat{\Omega}_\alpha = \hat{\sigma}_\alpha^2 I_{N_1}.$$
- **Case 2: Ω_α full block.** Let $\hat{\varepsilon}_{ijt}^{(\gamma,\lambda)}$ denote the residual from the within regression that partials out (γ_j, λ_t).
  $$\bar{\varepsilon}_{it} = \frac{1}{N_2} \sum_{j=1}^{N_2} \hat{\varepsilon}_{ijt}^{(\gamma,\lambda)}$$
  For each t, form the N_1-vector obtained by averaging over j element-by-element:
  $$\bar{\varepsilon}_{\cdot t} = (\bar{\varepsilon}_{1t}, \ldots, \bar{\varepsilon}_{N_1 t})'$$
  Then the base-block estimator is the average of outer products across t. Finally, to remove the contribution of u_{ijt} that remains after averaging over j, we subtract $\hat{\sigma}_u^2/N_2$ from the diagonal:
  $$\hat{\Omega}_\alpha = \frac{1}{T} \sum_{t=1}^{T} \bar{\varepsilon}_{\cdot t}\, \bar{\varepsilon}_{\cdot t}' - \frac{\hat{\sigma}_u^2}{N_2} I_{N_1}.$$

Analogous constructions are used for $\hat{\Omega}_\gamma$ and $\hat{\Omega}_\lambda$, with the differencing scheme rotated to isolate the relevant dimension.

Note: The fixed effect vectors are assumed to be drawn only once in case a model with fixed effects is estimated. Otherwise, the redraw structure for each of the random effect vectors is as follows:

- α: Redrawn T times, duplicated N_2 times.
- γ: Redrawn T times, duplicated $N1$ times.
- λ: Redrawn N_2 times, duplicated N_1 times.

4. Assemble the estimated covariance matrix

$$\hat{\Omega} = S_\alpha \, \hat{\Omega}_\alpha \, S'_\alpha + S_\gamma \, \hat{\Omega}_\gamma \, S'_\gamma + S_\lambda \, \hat{\Omega}_\lambda \, S'_\lambda + \hat{\sigma}_u^2 I_N.$$

5. Compute the GLS estimator

$$\hat{\beta}_{\text{FGLS2}} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}y.$$

(v) GLS. GLS is constructed in a similar fashion using known covariance matrices, and σ_u.

7.1.4 Monte Carlo Procedure

For each parameter configuration:

1. Draw $(\mathbf{\Omega}_\alpha, \mathbf{\Omega}_\gamma, \mathbf{\Omega}_\lambda)$.
2. For each replication:
   (a) Draw $\mathbf{X}$, $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$, $\boldsymbol{\lambda}$, and $\mathbf{u}$.
   (b) Construct $\mathbf{y}$ using (22).
   (c) Compute all estimators.
   (d) Store point estimates and estimated variance matrices.

7.2 Specifications.

1. **All homoskedastic.** $\alpha_i + \gamma_j + \lambda_t$, all `draw_once`

$$\mathbf{\Omega}_\alpha = \sigma_\alpha^2 \mathbf{I}_{N_1}, \quad \mathbf{\Omega}_\gamma = \sigma_\gamma^2 \mathbf{I}_{N_2}, \quad \mathbf{\Omega}_\lambda = \sigma_\lambda^2 \mathbf{I}_T.$$

2. **All homoskedastic.** $\alpha_{it} + \gamma_{jt} + \lambda_t$, i is redrawn.

$$\mathbf{\Omega}_\alpha = \sigma_\alpha^2 \mathbf{I}_{N_1} \otimes \mathbf{I}_T, \quad \mathbf{\Omega}_\gamma = \sigma_\gamma^2 \mathbf{I}_{N_2} \otimes \mathbf{I}_T, \quad \mathbf{\Omega}_\lambda = \sigma_\lambda^2 \mathbf{I}_T.$$

3. **i full block j and t homoskedastic; i uses mixed redraw; j and t draw-once.** $\alpha_{it} + \gamma_j + \lambda_t$

$$\mathbf{\Omega}_\alpha \ = \mathbf{\Omega}_{\alpha,\text{block}} \otimes I_T, \quad \mathbf{\Omega}_\gamma = \sigma_\gamma^2 \mathbf{I}_{N2}, \quad \mathbf{\Omega}_\lambda = \sigma_\lambda^2 \mathbf{I}_T,$$

with i draw mode `:mixed` and (j,t) draw modes `:draw_once`.

Mixed redraw: T vectors of size N1 are drawn from MVN, they are then repeated identically across j (so that later the residuals later differ across j only by the idiosyncratic noise). For the covariance matrix: If fixed then $S_\alpha \ = \ \mathbf{1}_T \otimes \mathbf{1}_{N_2} \otimes I_{N_1}$, and if redrawn then $S_\alpha \ = \ I_T \otimes \mathbf{1}_{N_2} \otimes I_{N_1}$. Omega is repeated across the diagonal, to build a matrix with the correct size.

4. **i full block j and t homoskedastic; i and j use mixed redraw; t draw-once.** $\alpha_{it} + \gamma_{jt} + \lambda_t$

$$\mathbf{\Omega}_\alpha \ = \mathbf{\Omega}_{\alpha,\text{block}} \otimes I_T, \quad \mathbf{\Omega}_\gamma \ = \sigma_\gamma \, \mathbf{I}_{N2} \otimes \mathbf{I}_T, \quad \mathbf{\Omega}_\lambda = \sigma_\lambda^2 \mathbf{I}_T,$$

with i, j draw mode `:mixed` and t draw modes `:draw_once`.

To do: Remove 3 and 4, and make 6 omega gamma block (only T goes to infinity also fixed)

7.3 Results

8 Garbage collection

In this case,

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \left(\hat{\beta}_0(\mathbf{M}_X), \hat{\boldsymbol{\beta}}_1(\mathbf{M}_1)'\right)'$$

where $\mathbf{M}_1 = \mathbf{I}_N - N^{-1}\mathbf{i}_N\mathbf{i}_N'$ and $\mathbf{M}_X = \mathbf{I}_N - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, , $\tilde{\mathbf{X}} = \mathbf{M}_1\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{i}_N \otimes \boldsymbol{\mu}_{X_1}'$. Moreover, $\hat{\boldsymbol{\beta}}_1(\mathbf{M}_1) - \boldsymbol{\beta}_1 = o_p(1)$.

Appendix A: Technical Proofs

Proof of Proposition 1: It is straightforward to show that

$$\hat{\boldsymbol{\beta}}(\mathbf{I}) = \boldsymbol{\beta}_0 + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{S}_\alpha\boldsymbol{\alpha} + \mathbf{S}_\gamma\boldsymbol{\gamma} + \mathbf{S}_\lambda\boldsymbol{\lambda}) + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \tag{A.1}$$

Under the standard assumptions of $\mathbf{X}$ and $\boldsymbol{\varepsilon}$, $\mathbf{X}'\mathbf{X} - \mathbf{Q}_X = o_p(1)$ and $\mathbf{X}'\boldsymbol{\varepsilon} = o_p(1)$ for some finite matrix $\mathbf{Q}_X$. Now consider $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{S}_\alpha\boldsymbol{\alpha}$ and under the assumptions on $\mathbf{X}$ and the fact that $\boldsymbol{\alpha}$ is a non-random vector which infinite sum converges to $\bar{\alpha}$, this gives

$$
\begin{aligned}
\left(\mathbf{X}'\mathbf{X}\right)^{-1} N^{-1}\mathbf{X}'\mathbf{S}_\alpha\boldsymbol{\alpha} =& \mathbf{Q}_X^{-1}(\boldsymbol{\mu}_X \otimes \mathbf{i}_N')\mathbf{S}_\alpha\boldsymbol{\alpha} + o_p(1) \\
=& N^{-1}\mathbf{Q}_X^{-1}(\boldsymbol{\mu}_X \otimes \mathbf{i}_N')(\mathbf{I}_{N_1} \otimes \mathbf{i}_{N_2} \otimes \mathbf{i}_T)\boldsymbol{\alpha} + o_p(1) \\
=& N^{-1}\mathbf{Q}_X^{-1}(\boldsymbol{\mu}_X \otimes \mathbf{i}_N \otimes N_2 \otimes T)\boldsymbol{\alpha} + o_p(1) \\
=& \bar{\alpha}\mathbf{Q}_X^{-1}\boldsymbol{\mu}_X + o_p(1).
\end{aligned}
$$

Similar arguments apply to $\mathbf{S}_\gamma\boldsymbol{\gamma}$ and $\mathbf{S}_\lambda\boldsymbol{\lambda}$. This completes the proof. ■

Proof of Proposition 2: The first part of the result is a direct application of the Frisch-Waugh-Lovell Theorem, see for example Frisch and Waugh (1933). Direct calcualation yields

$$
\hat{\beta}_0 = \beta_0 + \left(\mathbf{i}'\mathbf{M}_{X_1}\mathbf{i}\right)^{-1}\mathbf{i}'\mathbf{M}_{X_1}\left(\mathbf{S}_\alpha\boldsymbol{\alpha} + \mathbf{S}_\gamma\boldsymbol{\gamma} + \mathbf{S}_\lambda\boldsymbol{\lambda}\right) + \left(\mathbf{i}'\mathbf{M}_{X_1}\mathbf{i}\right)^{-1}\mathbf{i}'\mathbf{M}_{X_1}\boldsymbol{\varepsilon}. \tag{A.2}
$$

Since $\mathbf{M}_{X_1}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}$ under the orthogonal assumption between the regressors and $\boldsymbol{\varepsilon}$, the last term vanishes asymptotically. For the middle term, consider

$$
\begin{aligned}
\mathbf{i}'\mathbf{M}_{X_1}\mathbf{S}_\alpha\boldsymbol{\alpha} =& \mathbf{i}'\mathbf{S}_\alpha\boldsymbol{\alpha} - \mathbf{i}'\mathbf{X}_1\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{S}_\alpha\boldsymbol{\alpha} \\
=& \bar{\alpha} - \boldsymbol{\mu}_X'\mathbf{Q}_X^{-1}\boldsymbol{\mu}_X\bar{\alpha} + o_p(1) \\
=& \bar{\alpha}\left(1 - \boldsymbol{\mu}_X'\mathbf{Q}_X^{-1}\boldsymbol{\mu}_X\right) + o_p(1).
\end{aligned}
$$

The last line follows from the similar arguments as in the proof in Proposition 1. Apply the same arguments to the remaining terms i.e., $\mathbf{S}_\gamma\boldsymbol{\gamma}$ and $\mathbf{S}_\lambda\boldsymbol{\lambda}$ gives the second result. Note that $\mathbf{M}_1$ is a de-mean operator and thus $\hat{\boldsymbol{\beta}}(\mathbf{M}_1)$ is the OLS estimator for

$$
\tilde{y} = \tilde{X}\boldsymbol{\beta}_1 + \mathbf{W}\boldsymbol{\delta} + \varepsilon
$$

where $\tilde{y} = \mathbf{y} - \bar{y}\mathbf{i}$ and $\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{i}_N \otimes \boldsymbol{\mu}_X'$. Apply Proposition 1 and combine the fact that the mean of $\tilde{\mathbf{X}} = \mathbf{0}$ gives the third result. This completes the proof. ■

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